Landon Rutledge

Landon.rutledge2222@gmail.com | (804) 393-3455 | Portfolio: https://landonr22.github.io/

Work Experience

Research Assistant in Infrastructure Risk and Resilience, University College London (UCL) May 2025 – Present

- Developing a GIS-based multi-hazard risk assessment toolkit to identify critical corridors and nodes across the UK's multi-modal freight network, supporting prioritization of resilience interventions for flooding, landslides, and multi-hazard vulnerabilities
- Collaborating with a team of researchers and an external stakeholder at the British Geological Survey to align hazard modeling and outputs with real-world transportation adaptation needs
- Synthesizing research findings into clear documentation and technical summaries to support internal decision-making and ongoing tool development

GIS Technician I, ASRC Federal May 2023 – September 2024

- Supported maintenance and enhancement of live GIS datasets in a production environment, working with street centerlines, land parcels, and related spatial layers
- Reviewed, edited, and validated spatial features and attribute data using ArcGIS Pro and web tools (ArcGIS Online) to ensure consistency and positional accuracy
- Conducted parcel and address-level QA/QC by integrating county and locally sourced GIS data, progressing into a peer review role verifying coworkers' submissions before database updates

Education

University College London (UCL), London, United Kingdom

- Master of Science in Geospatial Information Science and Computing
- Grade: Highest academic honor of Distinction; Class rank: Fourth in class

James Madison University, Harrisonburg, VA

- Bachelor of Science in Geographic Science
- Minor: Environmental Science

Research Projects

Dissertation: ***Multi-hazard exposure, rerouting, and disruption costs for road freight on England's Strategic Road Network***

- Conducted a GIS-based risk assessment to identify road freight corridors in England that are particularly vulnerable to flooding and landslides
- Developed a probabilistic impact model in Python using Monte Carlo simulation to quantify disruption costs across flooding, landslide, and multi-hazard scenarios
- This research project is a subset of the broader DARe Hub multi-modal freight resilience initiative, contributing a focused analysis on road freight vulnerability and continuity planning

DARe Hub Research Assignment: ***Improving the operational resilience of multi-modal freight networks***

- Conducting spatial and network-based risk analysis to assess multi-hazard exposure and disruption impacts across road and rail freight infrastructure using climate and infrastructure datasets
- Modeling operational and financial disruptions to both rail and road freight using Python
- Deliverable includes a live-updating GIS-based StoryMap dashboard integrating weather and network disruption data to support freight resilience planning and operational decision-making

Volunteer Project: ***Human Spatial Cognition and Indoor Trajectories***

- Collaborated on a small-group research project analyzing real-world human movement patterns in indoor spaces
- Explored correlations between movement and sociodemographic factors
- Utilized QGIS to visualize, analyze, and interpret walking trajectories within indoor environments

Skills

- **Programming Languages:** Python, R, SQL, JavaScript
- **Databases:** PostgreSQL, Microsoft Access
- **Software:** ArcGIS Pro, QGIS, Git/GitHub, FME, ArcGIS Online, Business Analyst, Esri Model Builder, Survey123, Field Maps, MATLAB, Adobe Acrobat, Erdas Imagine, Microsoft Office

Academic Honors

- Hart Prize for Geospatial Sciences Award Winner, 2025 (UCL)
- President's List: Spring 2023; Dean's List: Spring 2020, Spring 2022, Fall 2022
- Member & Treasurer, Gamma Theta Upsilon (Geographical Honor Society) (January 2022 – May 2023)

Publications and Presentations

- *Improving the Operational Resilience of Multi-Modal Freight Networks*. Poster presented at:
    - Transportation Research Symposium 2025 – Rotterdam, Netherlands
- *Multi-Modal Freight Vulnerability to Water-Related Disruptions.* Poster presented at:
    - DARe Researcher Conference 2025 – Newcastle, England

Relevant Coursework (UCL)

Geospatial Science

- Conducted a GIS-based assessment of historical sea level change across England by processing tide gauge records and generating interpolated coastal trend surfaces in ArcGIS Pro
- Integrated elevation data (OS Terrain 50) and raster-to-vector workflows to delineate flood-prone zones (0-10 m) and assess spatial exposure
- Intersected flood-prone areas with building footprints to quantify and map 14,000+ vulnerable structures in Liverpool, producing a technical poster emphasizing spatial methods, data accuracy, and cartographic design

Spatial Analysis & Geocomputation

- Applied exploratory spatial data analysis (ESDA), including Local Moran's I, to identify clusters of high socio-economic deprivation and prioritize candidate local authority districts for further analysis
- Designed and implemented a multi-criteria decision analysis (MCDA) framework integrating land use, slope, proximity to rail stations and major roads, and brownfield site availability to evaluate site suitability
- Developed weighted overlay models in R and ArcGIS Pro to compare alternatives, quantify tradeoffs, and produce transparent suitability maps supporting evidence-based site selection under environmental and infrastructure constraints

Spatial Databases & Data Management

- Designed and implemented a relational PostgreSQL spatial database from scratch, including table schemas, constraints, views, and nested SQL queries to model and monitor hierarchical asset conditions
- Developed SQL-based decision rules and threshold logic to evaluate asset maintenance needs, supporting condition-based interventions and data-driven management decisions
- Integrated the spatial database with QGIS and FME to visualize, query, and validate geocoded assets, simulating real-world infrastructure asset management workflows